ITESO

Departamento de Electrónica, Sistemas e Informática

Big Data

Spring 2026

Lab 04: Data Unions and Joins Pipeline

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In [1]:
```
from spark_utils import SparkUtils

# Create Spark session
su = SparkUtils("Lab04", "spark://spark-master:7077")
```

```
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(ne
wLevel).
26/03/03 02:18:27 WARN NativeCodeLoader: Unable to load native-hadoop library for yo
ur platform... using builtin-java classes where applicable
```

In [9]:
```
!pwd
```

```
/opt/spark/work-dir/notebooks/labs
```

In [13]:
```
# 1. Agencies
agency_schema = SparkUtils.generate_schema([("agency_id", "int"), ("agency_info", "
df_agencies = su._spark.read.csv("/opt/spark/work-dir/data/car_service/agencies/",
df_agencies.show()

# 2. Brands
brand_schema = SparkUtils.generate_schema([("brand_id", "int"), ("brand_info", "str
df_brands = su._spark.read.csv("/opt/spark/work-dir/data/car_service/brands/", sche
df_brands.show()

# 3. Cars
car_schema = SparkUtils.generate_schema([("car_id", "int"), ("car_info", "string")]
df_cars = su._spark.read.csv("/opt/spark/work-dir/data/car_service/cars/", schema=c
```

```
df_cars.show()

# 4. Customers
customer_schema = SparkUtils.generate_schema([("customer_id", "int"), ("customer_in
df_customers = su._spark.read.csv("/opt/spark/work-dir/data/car_service/customers/"
df_customers.show()

# 5. Rentals
rental_schema = SparkUtils.generate_schema([("rental_id", "int"), ("rental_info", "
df_rentals = su._spark.read.csv("/opt/spark/work-dir/data/car_service/rentals/", sc
df_rentals.show()
```

```
+---------+--------------------+
|agency_id|         agency_info|
+---------+--------------------+
|        1|{'agency_name': '...|
|        2|{'agency_name': '...|
|        3|{'agency_name': '...|
|        4|{'agency_name': '...|
|        5|{'agency_name': '...|
+---------+--------------------+

+--------+--------------------+
|brand_id|          brand_info|
+--------+--------------------+
|       1|{'brand_name': 'M...|
|       2|{'brand_name': 'B...|
|       3|{'brand_name': 'A...|
|       4|{'brand_name': 'F...|
|       5|{'brand_name': 'B...|
|       6|{'brand_name': 'H...|
|       7|{'brand_name': 'T...|
+--------+--------------------+

+------+--------------------+
|car_id|            car_info|
+------+--------------------+
|     1|{'car_name': 'Cha...|
|     2|{'car_name': 'She...|
|     3|{'car_name': 'Fau...|
|     4|{'car_name': 'Wag...|
|     5|{'car_name': 'Cam...|
|     6|{'car_name': 'Arc...|
|     7|{'car_name': 'Pat...|
|     8|{'car_name': 'Jon...|
|     9|{'car_name': 'Gar...|
|    10|{'car_name': 'Lev...|
|    11|{'car_name': 'Gri...|
|    12|{'car_name': 'Moo...|
|    13|{'car_name': 'Sum...|
|    14|{'car_name': 'Cla...|
|    15|{'car_name': 'Ngu...|
|    16|{'car_name': 'Wal...|
|    17|{'car_name': 'Wes...|
|    18|{'car_name': 'Bol...|
|    19|{'car_name': 'Man...|
|    20|{'car_name': 'Har...|
+------+--------------------+
only showing top 20 rows
+-----------+--------------------+
|customer_id|       customer_info|
+-----------+--------------------+
|          1|{'customer_name':...|
|          2|{'customer_name':...|
|          3|{'customer_name':...|
|          4|{'customer_name':...|
|          5|{'customer_name':...|
|          6|{'customer_name':...|
```

```
|          7|{'customer_name':...|
|          8|{'customer_name':...|
|          9|{'customer_name':...|
|         10|{'customer_name':...|
|         11|{'customer_name':...|
|         12|{'customer_name':...|
|         13|{'customer_name':...|
|         14|{'customer_name':...|
|         15|{'customer_name':...|
|         16|{'customer_name':...|
|         17|{'customer_name':...|
|         18|{'customer_name':...|
|         19|{'customer_name':...|
|         20|{'customer_name':...|
+-----------+--------------------+
only showing top 20 rows
+---------+--------------------+
|rental_id|         rental_info|
+---------+--------------------+
|    11891|{'car_id': 21, 'c...|
|    11892|{'car_id': 11, 'c...|
|    11893|{'car_id': 22, 'c...|
|    11894|{'car_id': 5, 'cu...|
|    11895|{'car_id': 4, 'cu...|
|    11896|{'car_id': 8, 'cu...|
|    11897|{'car_id': 23, 'c...|
|    11898|{'car_id': 24, 'c...|
|    11899|{'car_id': 27, 'c...|
|    11900|{'car_id': 3, 'cu...|
|    11901|{'car_id': 3, 'cu...|
|    11902|{'car_id': 3, 'cu...|
|    11903|{'car_id': 29, 'c...|
|    11904|{'car_id': 3, 'cu...|
|    11905|{'car_id': 27, 'c...|
|    11906|{'car_id': 12, 'c...|
|    11907|{'car_id': 25, 'c...|
|    11908|{'car_id': 14, 'c...|
|    11909|{'car_id': 24, 'c...|
|    11910|{'car_id': 24, 'c...|
+---------+--------------------+
only showing top 20 rows
```

Extract a JSON column with get_json_object function

In [27]:

```
from pyspark.sql.functions import get_json_object

# Agencies
df_json_agencies = df_agencies.withColumn("agency_name", get_json_object("agency_in
df_json_agencies.show()

# Brands
df_json_brands = df_brands.withColumn("brand_name", get_json_object("brand_info", "
df_json_brands.show()

# Cars
```

```
df_json_cars = df_cars.withColumn("car_name", get_json_object("car_info", "$.car_na
                      .withColumn("brand_id", get_json_object("car_info", "$.brand_
df_json_cars.show()

# Customers
df_json_customers = df_customers.withColumn("customer_name", get_json_object("custo
df_json_customers.show()

# Rentals
df_json_rentals = df_rentals.withColumn("car_id", get_json_object("rental_info", "$
                        .withColumn("customer_id", get_json_object("rental_info", "
                        .withColumn("agency_id", get_json_object("rental_info", "$.
df_json_rentals.show()
```

```
+---------+--------------------+-------------+
|agency_id|         agency_info|  agency_name|
+---------+--------------------+-------------+
|        1|{'agency_name': '...|  NYC Rentals|
|        2|{'agency_name': '...|LA Car Rental|
|        3|{'agency_name': '...| Zapopan Auto|
|        4|{'agency_name': '...|      SF Cars|
|        5|{'agency_name': '...|  Mexico Cars|
+---------+--------------------+-------------+

+--------+--------------------+-------------+
|brand_id|          brand_info|   brand_name|
+--------+--------------------+-------------+
|       1|{'brand_name': 'M...|Mercedes-Benz|
|       2|{'brand_name': 'B...|          BMW|
|       3|{'brand_name': 'A...|         Audi|
|       4|{'brand_name': 'F...|         Ford|
|       5|{'brand_name': 'B...|          BYD|
|       6|{'brand_name': 'H...|        Honda|
|       7|{'brand_name': 'T...|       Toyota|
+--------+--------------------+-------------+

+------+--------------------+--------------------+--------+
|car_id|            car_info|            car_name|brand_id|
+------+--------------------+--------------------+--------+
|     1|{'car_name': 'Cha...|Chang-Fisher Model 7|       5|
|     2|{'car_name': 'She...|Sheppard-Tucker M...|       6|
|     3|{'car_name': 'Fau...|Faulkner-Howard M...|       3|
|     4|{'car_name': 'Wag...|  Wagner LLC Model 1|       5|
|     5|{'car_name': 'Cam...|  Campos PLC Model 4|       4|
|     6|{'car_name': 'Arc...|Archer-Patel Model 2|       3|
|     7|{'car_name': 'Pat...|Patrick, Barrera ...|       6|
|     8|{'car_name': 'Jon...|Jones, Jefferson ...|       2|
|     9|{'car_name': 'Gar...|Garcia, Hamilton ...|       6|
|    10|{'car_name': 'Lev...|  Levy Group Model 9|       3|
|    11|{'car_name': 'Gri...|Grimes-Green Model 8|       3|
|    12|{'car_name': 'Moo...|Moon, Davis and L...|       6|
|    13|{'car_name': 'Sum...|Summers, Barnett ...|       4|
|    14|{'car_name': 'Cla...|Clayton-Cook Mode...|       6|
|    15|{'car_name': 'Ngu...|Nguyen, Harrell a...|       2|
|    16|{'car_name': 'Wal...|Walker, Pratt and...|       6|
|    17|{'car_name': 'Wes...|West, Miller and ...|       1|
|    18|{'car_name': 'Bol...|Bolton, Burns and...|       6|
|    19|{'car_name': 'Man...| Mann-Kelley Model 3|       2|
|    20|{'car_name': 'Har...|Harris, Lloyd and...|       6|
+------+--------------------+--------------------+--------+
only showing top 20 rows
+-----------+--------------------+--------------------+
|customer_id|       customer_info|       customer_name|
+-----------+--------------------+--------------------+
|          1|{'customer_name':...|       Tiffany Riley|
|          2|{'customer_name':...|      Matthew Davies|
|          3|{'customer_name':...|      Rebecca Miller|
|          4|{'customer_name':...|       Katelyn Mccoy|
|          5|{'customer_name':...|         Dana Dennis|
|          6|{'customer_name':...|       Daniel Norton|
```

```
|          7|{'customer_name':...|       Robert Garcia|
|          8|{'customer_name':...|    Michael Williams|
|          9|{'customer_name':...|      Susan Ferguson|
|         10|{'customer_name':...|         Alyssa Reid|
|         11|{'customer_name':...|         Laura Myers|
|         12|{'customer_name':...|     William Johnson|
|         13|{'customer_name':...|         Tyler Lopez|
|         14|{'customer_name':...|         Daniel Owen|
|         15|{'customer_name':...|     Allison Coleman|
|         16|{'customer_name':...|        David Walker|
|         17|{'customer_name':...|        Hayley White|
|         18|{'customer_name':...|       Amanda Garcia|
|         19|{'customer_name':...|Christina Howard PhD|
|         20|{'customer_name':...|      Tamara Jackson|
+-----------+--------------------+--------------------+
only showing top 20 rows
+---------+--------------------+------+-----------+---------+
|rental_id|         rental_info|car_id|customer_id|agency_id|
+---------+--------------------+------+-----------+---------+
|    11891|{'car_id': 21, 'c...|    21|         71|        1|
|    11892|{'car_id': 11, 'c...|    11|         52|        2|
|    11893|{'car_id': 22, 'c...|    22|        116|        4|
|    11894|{'car_id': 5, 'cu...|     5|        107|        1|
|    11895|{'car_id': 4, 'cu...|     4|         53|        4|
|    11896|{'car_id': 8, 'cu...|     8|        131|        2|
|    11897|{'car_id': 23, 'c...|    23|         66|        3|
|    11898|{'car_id': 24, 'c...|    24|         60|        4|
|    11899|{'car_id': 27, 'c...|    27|         92|        2|
|    11900|{'car_id': 3, 'cu...|     3|         40|        4|
|    11901|{'car_id': 3, 'cu...|     3|         35|        1|
|    11902|{'car_id': 3, 'cu...|     3|         72|        1|
|    11903|{'car_id': 29, 'c...|    29|         58|        3|
|    11904|{'car_id': 3, 'cu...|     3|        131|        4|
|    11905|{'car_id': 27, 'c...|    27|         10|        2|
|    11906|{'car_id': 12, 'c...|    12|         40|        1|
|    11907|{'car_id': 25, 'c...|    25|         53|        3|
|    11908|{'car_id': 14, 'c...|    14|        153|        4|
|    11909|{'car_id': 24, 'c...|    24|         96|        4|
|    11910|{'car_id': 24, 'c...|    24|        151|        4|
+---------+--------------------+------+-----------+---------+
only showing top 20 rows
```

JOIN

In [29]:

```python
from pyspark.sql.functions import col

final_df = df_json_rentals \
    .join(df_json_cars, on="car_id") \
    .join(df_json_brands, on="brand_id") \
    .join(df_json_agencies, on="agency_id") \
    .join(df_json_customers, on="customer_id") \
    .select("rental_id", "car_name", "agency_name", "customer_name") \
    .orderBy(col("rental_id"), ascending=True) \
    .where(col("rental_id") >= 12740)
```

```
final_df.show(5)
```

```
[Stage 67:>                                                         (0 + 1) / 2]
+---------+--------------------+-------------+----------------+
|rental_id|            car_name|  agency_name|   customer_name|
+---------+--------------------+-------------+----------------+
|    12740|Villanueva PLC Mo...| Zapopan Auto|    Andrew Gould|
|    12741|  Levy Group Model 9|      SF Cars| Barbara Sanders|
|    12742|Nguyen, Harrell a...|      SF Cars|Katherine Ibarra|
|    12743|Grimes-Green Model 8|LA Car Rental|    Taylor Perry|
|    12744|Jones, Jefferson ...|      SF Cars|  Regina Stewart|
+---------+--------------------+-------------+----------------+
only showing top 5 rows
```

In [40]:
```
su._spark.stop()
```